Macroeconomics I

Neoclassical Growth Model
with an application to Real Business Cycles

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What We Learn in This Chapter

- How to solve the Neoclassical growth model using the social planner and the descentralized competitive equilibrium.
- How to use the Hamiltonian to solve continuous time problems.
- How to use the phase diagram to learn about the transitional dynamics of the neoclassical growth model.
- What are the conditions that guarantee a balanced growth path.
- How to solve a standard Real Business Cycle Model.

References

- PhD Macrobook Ch. 4, Ch. 6 and Ch. 14.
- Acemoglu Ch. 7 and 8.
- Dirk Krueger Ch. 3 and 9.

Introduction

- We are going to build our first dynamic GE macroeconomic model.
- In particular, we will study the **Neoclassical growth model** (AKA Ramsey-Cass-Koopmans).
- In this model, savings rate are endogenous and respond to changes in the environment.
- The model is the building block for most of the more complex models out there.

Discrete Time: Neoclassical Growth Model

- **Environment**: No uncertainty; Single good that can be consumed or invested $Y_t = C_t + I_t$. No population growth (yet), $L_t = 1$. No TFP growth (yet).
- **Preferences:** $U(\{C_t\}_{t=0}^{\infty}) = \sum_{t=0}^{\infty} \beta^t u(C_t)$
- **Technology**: $Y_t = F(K_t, L_t) = F(k_t, 1) \equiv f(k_t)$ and $I_t = K_{t+1} - (1-\delta)K_t$.
  - Production function follows the same assumptions as in Solow.
- **Government:** None (yet).
- **Endowments**: Initial capital K_0.
- **Equilibrium concept:** Competitive.

Solution: Sequences $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ (everything now will be in per-capita terms).

Utility

- The utility function will follow the usual assumptions:
  - Time-separable, $u'(c) > 0$, $u''(c) < 0$, $\beta \in (0, 1)$, time-invariant, and satisfies the Inada conditions.
- In particular, we will assume that the model is populated by a **representative household**.
- Solving the individual problem is enough to get the aggregate consumption, aggregate savings, etc.
- A lot of macro models assume a *representative household*. What does this mean?

Digression: Representative Agent

- In general, households can be heterogeneous in many dimensions (income, wealth, preferences).
- If we are interested in studying aggregate variables, we must aggregate the decisions of all households $\Rightarrow$ Problem: aggregating individual decisions is hard.
- **Solution:** Assume the existence of a "representative" household that characterizes aggregate behavior with a single agent.
- We can do that whenever some form of **Gorman Aggregation Theorem** holds in the utility function.
  - **Gorman Aggregation Theorem:** If all households have the same marginal propensity to consume, aggregate demand depends only on aggregate wealth — the distribution is irrelevant.

→ Details

Solving the Model

- We already have assumptions about preferences and technology.
- **Ultimate goal:** Find equilibrium allocations (and prices). How to do it:
  (i) **Decentralized equilibrium**: find the price that equates the supply of capital/consumption with its demand.
  (ii) **Social planner**: Solve the problem of the **benevolent central planner** $\Rightarrow$ Also the optimal/efficient solution of the model.
- If the **Welfare Theorems hold**, the solutions to both problems are the same.
- **Social Planner's Problem**:
  - Maximizes utility given the technological and resource constraints of the economy (not subject to consumer budget constraint - but rather to the TOTAL resources of the economy).

Social Planner's Problem

- Planner chooses the allocation $\{k_{t+1}, c_t\}_{t=0}^{\infty}$ that maximizes the representative household's utility.

$$\max_{\{k_{t+1}\geq 0,\quad c_t\geq 0\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t) \tag{1}$$

$$s.t. \quad c_t + i_t \leq y_t = f(k_t) \quad \forall t; \tag{2}$$

$$k_{t+1} = i_t + (1-\delta)k_t \quad \forall t; \tag{3}$$

$$k_0 > 0 \text{ given}; \tag{4}$$

- Substitute the capital accumulation equation into i_t, so the resource constraint reads: $c_t + k_{t+1} \leq f(k_t) + (1-\delta)k_t$.

Solving a Dynamic Problem

- We will first solve it with a finite T using constraint optimization methods (Kuhn-Tucker).
- The Kuhn-Tucker conditions are sufficient if the objective function is concave and the constraints are convex.
- The assumptions we made about u and f guarantee that these conditions are satisfied:
  - $u(c_t)$ is increasing so the resource constraint remains with equality.
  - $u(c_t)$ is concave, so the sum of $u(c_t)$ is also concave.
  - The constraint is convex: $0 \leq k_t \leq f(k_t) + (1 - \delta)k_t$.
  - Inada conditions ensure an interior solution $c > 0$ and $k > 0$.
  - Except for the last period T where $k_{T+1} = 0$.

Neoclassical Growth Model

Lagrangian:

$$\mathcal{L} = \sum_{t=0}^{T} \left[\beta^t u(c_t) + \lambda_t \left(f(k_t) + (1-\delta)k_t - c_t - k_{t+1}\right) + \mu_t k_{t+1}\right] \tag{5}$$

- Kuhn-Tucker conditions (for all t):
  - $k_{t+1} \geq 0$, $\lambda_t \geq 0$ and $\mu_t \geq 0$.
  - Complementary slackness: $k_{t+1}\mu_t = 0$

First-order conditions...

- Note that $k_{t+1} > 0$ and $\mu_t = 0$, for all $t = 0, .., T-1$:

$$u'(c_t)\beta^t = \lambda_t \quad \text{and} \quad \lambda_t = [f'(k_{t+1}) + 1 - \delta]\lambda_{t+1} \quad t = 0, ..., T-1 \tag{6}$$

- And we find the *Euler Equation*:

$$u'(c_t) = (f'(k_{t+1}) + 1 - \delta)\beta u'(c_{t+1}) \quad t = 0, 1, ..., T-1 \tag{7}$$

Euler Equation

- The Euler Equation the trade-off between consumption and saving (or in the case of the planner allocating one unit in c_t or in i_t).

$$\underbrace{u'(c_t)}_{\text{mg. cost of invest}} = \underbrace{(f'(k_{t+1}) + 1 - \delta)}_{\text{return on investment}} \underbrace{\beta u'(c_{t+1})}_{\text{mg. utility of consuming more in } t+1} \tag{8}$$

- Marginal cost of foregoing one unit of the final good in t is equal to the discounted marginal benefit of consuming $f'(k_{t+1}) + 1 - \delta$ units of the final good in $t+1$.

- Strict concavity in the utility function implies households would like to smooth consumption over the lifetime.

- Note that extra saving changes future returns via $f'(k_{t+1}) + 1 - \delta$.

Solving the Problem: Finite Time

- Note that the Euler equation is only valid until period $T-1$. The FOCs in period T:

$$u'(c_T)\beta^T = \lambda_T \quad \text{and} \quad \lambda_T = \mu_T \quad t = 0, ..., T-1 \tag{9}$$

- This implies that $\mu_T = \lambda_T > 0$ and, since $k_{T+1}\mu_T = 0$, $k_{T+1} = 0$.

- Intuitive result, since it makes no sense to take capital to $T+1$.

Finite Time Solution

- Utility-maximizing sequences must satisfy the system of difference equations (for $t = 0, ..., T-1$):

$$u'(c_t) = (f'(k_{t+1}) + 1 - \delta)\beta u'(c_{t+1}) \quad \text{(Euler Equation)} \tag{10}$$

$$c_t + k_{t+1} = f(k_t) + (1-\delta)k_t \quad \text{(Resource Constraint)} \tag{11}$$

- Alternatively, we can substitute c_t and write the problem as a second-order difference equation.
- Two first-order difference equations require two conditions: an initial and a terminal condition.
- These conditions are: given by k_0 and $k_{T+1} = 0$.

Infinite Time

- In finite time $k_{T+1} = 0$. But in infinite time what is the terminal condition that ensures the system of equations has a unique solution? The *Transversality Condition* (TVC).
- Note that in finite time: $\lambda_T k_{T+1} = 0$.
- The **Transversality Condition**:

$$\lim_{T\to\infty} \lambda_T k_{T+1} = \lim_{T\to\infty} \beta^T u'(c_T) k_{T+1} = 0 \tag{12}$$

- Intuitively, it says that the shadow value of capital converges to zero (not the capital stock).
  - In this sense, it is not optimal for the planner to choose a sequence of capital involving a positive shadow value in present value (as it was not optimal to $k_{T+1} > 0$ in finite time).
- Without TVC, there exist infinite sequences satisfying the EE that are not optimal — specifically, *bubble paths* where capital grows without bound because the planner keeps investing instead of consuming optimally.
- Check the **Proof** for sufficiency of the TVC + EE in the PhD macrobook (Proposition 4.4).

Infinite Time

- With the two first-order difference equations (EE and resource constraint) and the initial and terminal condition (k_0 and TVC), we can find the optimal allocations that solve the central planner's problem.
- In most applications it is not possible to solve the problem analytically. You must:
  - Use linear approximations.
  - Solve the problem on the computer (using dynamic programming).
- There is one case you can solve the model analytically:
- **Example**: Suppose $u(c) = \log(c)$, $\delta = 1$, $f(k) = k^{\alpha}$ (i.e., $F()$ is Cobb-Douglas). Solve for the optimal policy, i.e. k_{t+1} as a function of k_t and the parameters.

→ Solution

Neoclassical Growth: Decentralized Equilibrium

- We already solved for optimal growth allocations $\Rightarrow$ Social Planner.
- We know that by the Welfare Theorems the allocations chosen by the planner are the same as the one of the competitive equilibrium (under certain conditions).
- Okay, but what about prices? And what if the Welfare Theorems don't apply?

Neoclassical Growth Model: Firms

- The representative firm's problem is the same as in the Solow Model:

$$\max_{K_t, L_t} \pi_t = F(K_t, L_t) - r_t K_t - w_t L_t \tag{13}$$

- Given the assumptions on $F(K_t, L_t)$, the FOCs are necessary and sufficient:

$$r = MPK = F_K(K, L) = f'(k) \tag{14}$$

$$w = MPL = F_L(K, L) = f(k) - f'(k)k \tag{15}$$

Neoclassical Growth Model

Households' Problem

- Households own the capital and rent it to the firms through an asset market (receiving net returns $r_t - \delta$).
- Households supply labor (inelastically) to firms.

$$\max_{\{a_{t+1},\ c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t) \tag{16}$$

$$s.t. \quad c_t + a_{t+1} \leq (1 + r_t - \delta) a_t + w_t \quad \forall t; \tag{17}$$

$$a_t \geq 0 \ \forall t \quad \text{and} \quad a_0 > 0 \text{ given}; \tag{18}$$

- Note that factor prices have a t-subscript. The budget constraint is already in per-capita terms (otherwise, labor income would be $w_t L_t$).

Neoclassical Growth Model

Households' Problem

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_t) + \lambda_t \left(w_t + (1 + r_t - \delta)a_t - a_{t+1} - c_t\right) \tag{19}$$

- Given the assumptions we made on F and u, we know that the solution will be interior and the budget constraint holds with equality.
- FOCs: $\beta^t u'(c_t) = \lambda_t$ and $\lambda_{t+1}(1 + r_{t+1} - \delta) = \lambda_t$ for all t.
- Solution of the problem is the sequence that satisfies the Euler Equation:

$$u'(c_t) = \beta(1 + r_{t+1} - \delta)u'(c_{t+1}) \quad \forall t \tag{20}$$

together with the Transversality Condition.

Households' Problem: TVC vs No-Ponzi Game

- Note that the TVC is related to the *no-Ponzi game*, but they are conceptually different.
- Using $a_t = k_t$ and the EE: $\lambda_t = \dfrac{\lambda_{t-1}}{(1 + r_t - \delta)}$, so:

$$\lambda_T = \frac{\lambda_0}{\prod_{t=0}^{T}(1 + r_t - \delta)} \tag{21}$$

- Substituting into TVC ($\lim_{T\to\infty} \lambda_T k_{T+1} = 0$) and since $\lambda_0 > 0$:

$$\lim_{T\to\infty} \frac{k_{T+1}}{\prod_{t=0}^{T}(1 + r_t - \delta)} = 0$$

  i.e., TVC *implies* the no-Ponzi condition holds with equality at the optimum.
- Although they seem to have the same utility, they are different things:
  - The *no-Ponzi game* is a restriction in the households' problem that prevents the accumulation of debt.
  - In the basic neoclassical growth model, the no-Ponzi condition is usually omitted since $a_t = k_t > 0 \; \forall t$. But in more sophisticated versions (with different types of bonds, government, etc.) it may be necessary.

TVC vs No-Ponzi Game

- TVC determines the optimal choice given a set of possible sequences.
- It is a necessary and sufficient condition for the solution of the problem in the **sequential formulation** of the growth model.
  - In other words, it is a **terminal condition**.
- **Kamihigashi (2008)**: "A no-Ponzi-game condition is a constraint that prevents overaccumulation of debt, while a typical transversality condition is an optimality condition that rules out overaccumulation of wealth. They place opposite restrictions, and should not be confused."
  - **No-Ponzi** (constraint — rules out debt explosion from below):
$$\lim_{T\to\infty} \frac{a_{T+1}}{\prod_{t=0}^{T}(1+r_t-\delta)} \geq 0$$
  - **TVC** (optimality — rules out over-accumulation from above):
$$\lim_{T\to\infty} \beta^T u'(c_T)\, a_{T+1} = 0$$

Neoclassical Growth Model: Equilibrium Conditions

- Market clearing for capital and labor:

$$L_t^d = 1 \quad \text{and} \quad k_t^d = a_t \quad \forall t \tag{22}$$

- Market clearing in the goods market (resource constraint):

$$y_t = c_t + i_t \quad \forall t \tag{23}$$

  which is trivially satisfied by the households' budget constraint: $y_t = f(k_t) = r_t k_t + w_t$ and $i_t = k_{t+1} - (1-\delta)k_t$.

- By Walras's Law with two markets in equilibrium, the third will also be. Note that we solve for two prices for every t: r_t and w_t (the price of the final good was normalized $p_t = 1$).

Neoclassical Growth Model: Competitive Equilibrium

Definition. A competitive equilibrium is a sequence of allocations $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ for the consumer and the firm $\{k_t^d, L_t^d\}_{t=0}^{\infty}$, and prices $\{w_t, r_t\}_{t=0}^{\infty}$ such that:

1. Given k_0 and the sequence of interest rates and wages $\{r_t, w_t\}_{t=0}^{\infty}$, $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ is the solution to the household's problem.
2. Given the sequence of interest rates and wages $\{r_t, w_t\}_{t=0}^{\infty}$, $\{k_t^d, L_t^d\}_{t=0}^{\infty}$ is the solution to the firm's problem.
3. The markets clear for every t:

$$
\begin{aligned}
L_t^d &= 1 \\
k_t^d &= a_t \\
f(k_t) &= c_t + k_{t+1} - (1-\delta)k_t
\end{aligned}
$$

Equilibrium in Neoclassical Growth

- Combining the household's solution (Euler equation + budget constraint) with the firm's solution (factor market price equal to marginal product), we have:

$$\begin{aligned} u'(c_t) &= \beta(1 + r_{t+1} - \delta)u'(c_{t+1}) \quad \forall t \\ c_t + k_{t+1} - (1-\delta)k_t &= r_t k_t + w_t = y_t = f(k_t) \quad \forall t \\ r_t &= F_K(K_t, L_t) = MPK \quad \forall t \\ w_t &= F_L(K_t, L_t) = MPL \quad \forall t \end{aligned}$$

- Resulting in the same system as the social planner's:

$$u'(c_t) = (f'(k_{t+1}) + 1 - \delta)\beta u'(c_{t+1}) \qquad \text{(Euler Equation)}$$

$$c_t + k_{t+1} = f(k_t) + (1-\delta)k_t \qquad \text{(Resource Constraint)}$$

- EE + resource constraint + k_0 + TVC characterize the equilibrium sequence $\{c_t, k_{t+1}\}_{t=0}^{\infty}$.

Steady State

- **Steady State**: An economy is in a steady state when its variables assume a constant value over time.

$$k_{ss} = k_{t+1} = k_t$$
$$c_{ss} = c_{t+1} = c_t.$$

- Given our assumptions - especially concavity of F and constant returns to scale - the economy will converge to a steady states (intuition comes from the Solow model).

Steady State

- For the economy to reach the steady state, it suffices to start with $k_0 > 0$.
- Note that using the EE: $u'(c_{ss}) = \beta(1 + r_{ss} - \delta)u'(c_{ss})$ together with $r_{ss} = f'(k_{ss})$ we can easily find k_{ss}.
  - If $k_0 < k_{ss}$, the economy will accumulate capital until it reaches the steady state.
  - If $k_0 > k_{ss}$, the economy will decumulate capital until it reaches the steady state.
- We will study the accumulation dynamics in more detail later.
- **Example:** Find k_{ss} given $f(k) = k^{\alpha}$.

Steady State

Steady State with $F(K, L) = K^\alpha L^{1-\alpha}$

$$u'(c_{ss}) = \beta(1 + r_{ss} - \delta)u'(c_{ss}) \tag{24}$$

$$c_{ss} + \delta k_{ss} = r_{ss}k_{ss} + w_{ss}L_{ss} \tag{25}$$

$$r_{ss} = \alpha \left(\frac{k_{ss}}{L_{ss}}\right)^{\alpha-1} \tag{26}$$

$$w_{ss} = (1-\alpha)\left(\frac{k_{ss}}{L_{ss}}\right)^{\alpha} \tag{27}$$

- Given that $L_{ss} = 1$, it is a system of 4 equations and 4 endogenous variables $\{k_{ss}, c_{ss}, r_{ss}, w_{ss}\}$.
- There is no dynamics, so it is possible to find the analytical solution for the endogenous variables.

Dynamic Optimization in Continuous Time

Introduction

- Discrete time:

$$\sum_{t=0}^{\infty} \beta^t u(c_t)$$

- Continuous time:

$$\int_0^{\infty} e^{-\rho t} u(c_t) dt$$

- Why is discounting exponential?
- What's the difference?
  - There is no substantial difference. Some problems are naturally written in discrete time, others in continuous time (e.g., optimal stopping time problems).
  - Mathematics tends to be more elegant but sometimes more complicated.
- How to solve the problem?

Intuition

- The discount rate between discrete and continuous time are equivalent: $\beta^t = \left(\frac{1}{1+\rho}\right)^t$.
- **Intuition:** Suppose a period of t years. We can calculate compound interest:

$$\left(\frac{1}{1+r/n}\right)^t \times \cdots \times \left(\frac{1}{1+r/n}\right)^t = \left(\frac{1}{1+r/n}\right)^{nt}$$

  - If $n = 1$, we use annual interest. If $n = 4$, quarterly interest, etc.
- In continuous time, $n \to \infty$:

$$\lim_{n\to\infty} \left(\frac{1}{1+r/n}\right)^{nt} = e^{-rt}$$

- **Proof:** define $s \equiv n/r$, take the limit $s \to \infty$, and use L'Hôpital's rule.

Some Important Tricks

Continuous Time Growth:

- Growth rate over an interval Δt:

$$\frac{x_{t+\Delta t} - x_t}{x_t \Delta t} \quad \text{or in logs} \quad \frac{\ln x_{t+\Delta t} - \ln x_t}{\Delta t}$$

  taking the limit:

$$\lim_{\Delta t \to 0} \frac{x_{t+\Delta t} - x_t}{x_t \Delta t} = \frac{\dot{x}_t}{x_t} = g$$

- Suppose a variable $x = X/L$, where X grows at rate g and L at rate n:

$$\frac{\dot{x}_t}{x_t} = \frac{\dot{X}_t}{X_t} - \frac{\dot{L}_t}{L_t} = g - n.$$

- **Log-derivative trick**: the growth rate of $X(t)$ equals the time derivative of its logarithm:

$$\frac{\dot{X}(t)}{X(t)} = \frac{d}{dt} \log X(t).$$

  Useful for products and ratios: e.g., $Y = K^\alpha L^{1-\alpha} \Rightarrow \dot{Y}/Y = \alpha \dot{K}/K + (1-\alpha)\dot{L}/L$.

Example: Consumption and Saving

- Finite continuous time: $t \in [0, T]$.

$$\max_{c_t} \int_0^T e^{-\rho t} u(c_t) dt + e^{-\rho T} V_T(a_T)$$
$$s.t. \quad \frac{\partial a_t}{\partial t} = \dot{a}_t = r a_t + w - c_t,$$
$$a_0 \text{ given and } a_T \geq 0.$$

  where $V_T(a_T)$ is a terminal value (exogenous).

- Alternatively, we can impose a condition: $a_T = 0$ (not an agent's choice, but a restriction in the problem!).
- The solutions c_t and a_t are functions of time: $c : [0, T] \to \mathbb{R}$.

Example: Consumption and Saving

- How to find the budget constraint in continuous time?
- Budget constraint for a period Δt:

$$a_{t+\Delta t} = (a_t + w\Delta t - c_t \Delta t)(1 + r\Delta t)$$

- Flow vs stock: a_t is stock and $c_t,\ w$, and r are flow variables.
- Rewriting and taking the limit $\Delta t \to 0$:

$$\frac{a_{t+\Delta t} - a_t}{\Delta t} = ra_t + w - c_t + (w - c_t)r\Delta t$$

- Note, if $a_{t+\Delta t} = a_t(1 + r\Delta t) + w\Delta t - c_t\Delta t$, solution will be the same (why?).

Continuous Time

- How to solve the problem?
  - Variational calculus (we won't see it).
  - Pontryagin's Maximum Principle (analogous to the Lagrangian).
  - Hamilton-Jacobi-Bellman equation (analogous to the Bellman equation, likely won't see it).
- The approach here will be more "intuitive" and less formal. We will skip most of the theorems and proofs.
- Intuitively, much of what we've seen for discrete time has an equivalent for continuous time (transversality condition, etc).
- Acemoglu's Chapter 7 is the reference if you are interested in more details.

Pontryagin's Maximum Principle

- Consider the general problem:

$$\max_{y_t, x_t} \int_0^T f(x_t, y_t, t)dt + M(x_T)$$
$$s.t. \quad \dot{x}_t = g(x_t, y_t, t)$$
$$x_0 \text{ given.}$$

- x_t is the state vector.
- y_t is the control vector.
- f is the return function (with implicit discounting).
- g is the state's law of motion.

Pontryagin's Maximum Principle

- Define the Hamiltonian as:

$$H(x_t, y_t, \lambda_t, t) = f(x_t, y_t, t) + \lambda_t g(x_t, y_t, t)$$

  where λ_t is the **costate**, which is of the same dimension as x_t, and is a function of time.

- **Pontryagin's Maximum Principle.** Suppose f and g are continuously differentiable and (x_t^*, y_t^*) are continuous interior solutions. Then, there exists a function λ_t^* that satisfies the necessary conditions:

$$H_x(x_t, y_t, \lambda_t, t) = -\dot{\lambda}_t^* \quad \forall t, \tag{28}$$

$$H_y(x_t, y_t, \lambda_t, t) = 0 \quad \forall t, \tag{29}$$

$$H_\lambda(x_t, y_t, \lambda_t, t) = \dot{x}_t \quad \forall t. \tag{30}$$

  With the terminal condition $\lambda_T = M_x(x_t)$ (if the terminal condition is $a_T = 0$, then $\lambda_T = 0$).

Pontryagin's Maximum Principle

- For some intuition, imagine the following Lagrangian:

$$\mathcal{L}(x_t, y_t, \lambda_t, t) = \int_0^T [\underbrace{f(x_t, y_t, t) + \lambda_t g(x_t, y_t, t)}_{H(x_t, y_t, \lambda_t, t)} - \lambda_t \dot{x}_t] + M(x_T)$$

- λ_t acts as the "multiplier" and informs about the value of relaxing the constraints.
- The necessary conditions are analogous to the first-order conditions of the Lagrangian plus a "temporal' condition.

Pontryagin's Maximum Principle

- When the return function is exponentially discounted by $e^{-\rho t}$ (practically all economics' problem), it is convenient to redefine the Hamiltonian in **current value** $\hat{H}(x_t, y_t, \mu_t, t)$:

$$
\begin{aligned}
H(x_t, y_t, \lambda_t, t) =& f(x_t, y_t, t) + \lambda_t g(x_t, y_t, t) \\
H(x_t, y_t, \lambda_t, t) =& e^{-\rho t} \underbrace{\left(\hat{f}(x_t, y_t, t) + \mu_t g(x_t, y_t, t)\right)}_{\hat{H}(x_t, y_t, \mu_t, t)}
\end{aligned}
$$

- Where the multiplier and the return function are in current values:

$$
\mu_t = \lambda_t e^{\rho t}, \text{ and } \hat{f}(x_t, y_t, t) = f(x_t, y_t, t) e^{\rho t}.
$$

- The only condition that changes is (28):

$$
\begin{aligned}
\hat{H}_x(x_t, y_t, \mu_t, t) &= \rho \mu_t - \dot{\mu}_t^* \\
\hat{H}_y(x_t, y_t, \mu_t, t) &= 0 \\
\hat{H}_\lambda(x_t, y_t, \mu_t, t) &= \dot{x}_t.
\end{aligned}
$$

Example: Consumption and Saving

- In the Consumption and Saving problem:

$$\hat{H} = u(c_t) + \mu_t(ra_t + w - c_t)$$

- Hence:

$$r\mu_t = \rho\mu_t - \dot{\mu}_t^* \quad \forall t, \tag{31}$$

$$u'(c_t) = \mu_t \quad \forall t. \tag{32}$$

- Taking the derivative with respect to time:

$$\dot{\mu}_t = u''(c_t)\dot{c}_t \tag{33}$$

- Substituting, we find the Euler's equation (in continuous time):

$$\frac{u''(c_t)\dot{c}_t}{u'(c_t)} = -(r - \rho) \tag{34}$$

Example: Consumption and Saving

- The solution is a system of ordinary differential equations:

$$\frac{u''(c_t)\dot{c}_t}{u'(c_t)} = -(r - \rho)$$
$$\dot{a}_t = ra_t + w - c_t$$

- Alternatively, we can solve for $\dot{c}_t$ (using $\ddot{a}_t$) and find a second-order differential equation.
- To characterize a solution, we need an initial and a terminal condition.
  - Initial condition: a_0 given.
  - Terminal condition: $V_T(a_T)$ or transversality in the case of infinite time.
- Note that the intertemporal elasticity of substitution is equal to

$$\frac{1}{\sigma} = -\frac{u'(c_t)}{u''(c_t)c_t}$$

where σ is the coefficient of risk aversion. Then, the Euler equation: $\dot{c}_t/c_t = (r-\rho)/\sigma$.

Digression: Solution of Differential Equations

- Suppose a first-order linear non-homogeneous differential equation:

$$\dot{y}(t) + g(t)y(t) = f(t),$$

    - where $g(t)$ and $f(t)$ are parameters that may or may not depend on t.
- The solution is given by the following formula:

$$y(t) = \frac{\int_{t_0}^{t} u(s)f(s)ds + C}{u(t)}$$

    - where $u(t)$ is the integration factor given by:

$$u(t) = \exp\left(\int_{t_0}^{t} g(s)ds\right),$$

    - t_0 the initial time and C an arbitrary constant (which can be found with an initial condition $y(0)$, or some other condition).

Example: Consumption and Saving

- Using the formulas, we can solve the EE (an ODE) $\dot{c}_t = c_t(r-\rho)/\sigma$:

$$g(t) \equiv \frac{\rho - r}{\sigma}, \qquad f(t) \equiv 0 \qquad \text{and} \qquad t_0 = 0.$$

- The integration factor is $u(t) = \exp\left(\int_0^t (\rho - r)/\sigma ds\right) = e^{\frac{(\rho-r)t}{\sigma}}$.
- Therefore, for an arbitrary constant κ_1, we have the solution:

$$c_t = e^{\frac{(r-\rho)t}{\sigma}} \kappa_1.$$

- When $t = 0$, we can find that the constant is equal to consumption in period 0: $\kappa_1 = c_0$.

Example: Consumption and Saving

- Substituting c_t in the budget constraint:

$$\dot{a}_t = ra_t + w - c_t \qquad \Rightarrow \qquad \dot{a}_t - ra_t = w - e^{\frac{(r-\rho)t}{\sigma}} c_0$$

- We can apply the formulas:

$$g(t) \equiv -r, \qquad f(t) \equiv w - e^{\frac{(\rho-r)t}{\sigma}} c_0 \qquad \text{and} \qquad u(t) = \exp\left(\int_0^t -r ds\right) = e^{-rt}.$$

- The optimal saving a_t for an arbitrary constant κ_2 is:

$$a_t = e^{rt}\left[\int_0^t e^{rs}\left(w - e^{\frac{(r-\rho)s}{\sigma}} c_0\right) ds + \kappa_2\right]$$

Example: Consumption and Saving

- Solving the integral (assume $\sigma = 1$ for simplicity):

$$a_t = e^{rt}\left[\int_0^t (we^{-rs} - c_0 e^{-\rho s})ds + \kappa_2\right] = e^{rt}\left[\frac{w}{r}(1 - e^{-rt}) - \frac{c_0}{\rho}(1 - e^{-\rho t}) + \kappa_2\right]$$

- Substituting $t = 0$, we find that the constant is equal to the initial condition: $\kappa_2 = a_0$.
- We still need c_0. Since we are in finite time, we use the final condition: $a_T = 0$:

$$\begin{aligned}
0 &= e^{rT}\left[\frac{w}{r}(1 - e^{-rT}) - \frac{c_0}{\rho}(1 - e^{-\rho T}) + a_0\right] \\
c_0 &= \frac{\rho}{1 - e^{-\rho T}} \quad \underbrace{\left[\frac{w}{r}(1 - e^{-rT}) + a_0\right]}_{\text{present value of permanent income}}
\end{aligned}$$

- In infinite time, we could use the *no-Ponzi* condition as a terminal condition.

Neoclassical Growth in Continuous Time: The Ramsey-Cass-Koopmans Model

Environment, Preferences and Technology

- Infinite horizon and continuous time. Representative household with instantaneous utility $u(c_t)$.
  - $u(c_t)$ strictly increasing, concave, twice differentiable, satisfies Inada conditions.
- **Demographics**: $L_0 = 1$ and population growth at rate n: $L_t = e^{nt}$.
- All households supply labor inelastically (i.e., no work-leisure decision).
- The firm's problem is static so it is the same as in discrete time:

$$r_t = F_K(K_t, L_t) = f'(k_t)$$
$$w_t = F_L(K_t, L_t) = f(k_t) - k_t f'(k_t)$$

Environment and Preferences

- Recall per-capita consumption: $c_t = \frac{C_t}{L_t}$, where C_t is aggregate consumption.
- Utility function:

$$\int_0^\infty e^{-\rho t} L_t u(c_t) dt = \int_0^\infty e^{-(\rho-n)t} u(c_t) dt$$

- Assumption to ensure bounded integral: $\rho > n$.

Household Problem

- The (aggregate) budget constraint is:

$$\dot{\mathcal{A}}_t = r_t \mathcal{A}_t + w_t L_t - C_t,$$

  where $\mathcal{A}_t$ is aggregate asset quantity. Define $a_t = \mathcal{A}_t / L_t$ and we have the per capita budget constraint:

$$\dot{a}_t = (r_t - \delta - n) a_t + w_t - c_t.$$

- As we have seen, equilibrium in the asset market $a_t = k_t$ (but not necessarily in models with government bonds or other risky assets).

- And the *no-Ponzi game* condition in continuous time:

$$\lim_{t \to \infty} a_t \exp\left(- \int_0^t (r_s - \delta - n) ds \right) \geq 0.$$

Household Problem

- The household's problem:

$$
\begin{aligned}
& \max_{c_t \geq 0} \int_0^\infty e^{-(\rho - n)t} u(c_t) dt \\
s.t. \quad & \dot{a}_t = (r_t - \delta - n) a_t + w_t - c_t, \\
& a_0 \text{ given}, \\
& \lim_{t \to \infty} a_t \exp\left(- \int_0^t (r_s - \delta - n) ds \right) \geq 0.
\end{aligned}
$$

Equilibrium

Definition: Competitive equilibrium (sequential) consists of allocations for the households $\{c_t, a_t\}_{t=0}^{\infty}$, allocations for the firms $\{K_t, L_t\}_{t=0}^{\infty}$, and prices $\{r_t, w_t\}_{t=0}^{\infty}$ where:

1. Given prices and $a_0 = K_0/L_0$, allocations $\{c_t, a_t\}_{t=0}^{\infty}$ solve the household's problem.
2. Given prices, allocations $\{K_t, L_t\}_{t=0}^{\infty}$ solve the firm's problem:

$$\max_{K_t, L_t} F(K_t, L_t) - r_t K_t - w_t L_t$$

3. Market clearing for the labor, capital, and goods markets.

$$\begin{aligned}
&e^{nt} L_0 = L_t \\
&a_t L_t = K_t \\
&F(K_t, L_t) = \dot{K}_t + \delta K_t + L_t c_t
\end{aligned}$$

Characterizing the Equilibrium

- The (current value) Hamiltonian of the HH's problem:

$$\hat{H}(a_t, c_t, \mu_t) = u(c_t) + \mu_t(a_t(r_t - \delta - n) + w_t - c_t)$$

- Necessary conditions (together with the no-Ponzi and state LOM):

$$\begin{aligned} u'(c_t) &= \mu_t \\ \mu_t(r_t - \delta - n) &= -\dot{\mu}_t + (\rho - n)\mu_t \end{aligned}$$

- Implying the Euler equation:

$$\frac{u''(c_t)\dot{c}_t}{u'(c_t)} = -(r_t - \delta - \rho).$$

or using the intertemporal substitution elasticity: $1/\sigma(c_t) = -u'(c_t)/(u''(c_t)c_t)$:

$$\frac{\dot{c}_t}{c_t} = \frac{(r_t - \delta - \rho)}{\sigma(c_t)}.$$

→ Discrete vs. Continuous EE

Characterizing the Equilibrium

- Substituting r_t:

$$\frac{\dot{c}_t}{c_t} = \frac{(f'(k_t) - \delta - \rho)}{\sigma(c_t)}.$$

- And the *market clearing* (derived from the budget constraint):

$$\dot{k}_t = f(k_t) - (\delta + n)k_t - c_t$$

- The solution $\{c_t, k_t\}_{t=0}^{\infty}$ is characterized by the system of differential equations, together with the initial/terminal conditions k_0 and TVC ($\lim_{T\to\infty} e^{-\rho T}\mu_T k_T = 0$).

- Note that welfare theorems are satisfied and the Planner's solution equals the decentralized equilibrium.

Steady State

- Steady state: variables are constant over time, $\dot{k}_t = 0$ and $\dot{c}_t = 0$.
- Via SS we can find k_{ss} as a function of f, ρ, and δ (does not depend on the form of the utility function!):

$$\underbrace{f'(k_{ss})}_{r_{ss}} - \delta = \rho > n$$

- Define the *Golden Rule* as the capital that maximizes consumption:

$$\frac{dc}{dk} = f'(k_{ss}) - (\delta + n) = 0$$

- That is, the capital chosen by the Planner **is lower** than the *Golden Rule*. This happens because the Planner considers that households discount future consumption.

Steady State

- Unlike the Solow model, in RCK k_{ss} does not depend on population growth!
  - Ramsey: $f'(k_{ss}) = \delta + \rho$,
  - Solow (cns time): $\dfrac{f(k_{ss})}{k_{ss}} = \dfrac{\delta + n}{s}$,

  Note the connection between the savings rate s in Solow and the discount rate in RCK $\Rightarrow\uparrow \rho$ more impatient and lower capital accumulation.
- Once we have k_{ss} computing the rest is easy:
  - Aggregate resource constraint: $c_{ss} = f(k_{ss}) - (\delta + n)k_{ss}$.
  - Savings rate:

$$c_{ss} = (1 - s_{ss})f(k_{ss}) \Leftrightarrow s_{ss} = \frac{(\delta + n)k_{ss}}{f(k_{ss})}$$

Steady State

- **Example**: Use $f(k_t) = Ak_t^\alpha$ and do comparative statics of the effects of A, δ, n, and ρ on c_{ss} and k_{ss}.

$$k_{ss} = \left(\frac{\alpha A}{\delta + \rho}\right)^{1/(1-\alpha)}$$

$$c_{ss} = k_{ss}(Ak_{ss}^{\alpha-1} - (\delta + n)) = k_{ss}\left(\frac{(\delta+\rho) - \alpha(\delta+n)}{\alpha}\right)$$

- Since $\rho > n$ and $\alpha < 1$, consumption in the SS is a fraction of capital in the SS.
  - $\partial k_{ss}/\partial A > 0$ and $\partial c_{ss}/\partial A > 0$
  - $\partial k_{ss}/\partial \rho < 0$ and $\partial c_{ss}/\partial \rho < 0$
  - $\partial k_{ss}/\partial \delta < 0$ and $\partial c_{ss}/\partial \delta < 0$
  - $\partial k_{ss}/\partial n = 0$ and $\partial c_{ss}/\partial n < 0$

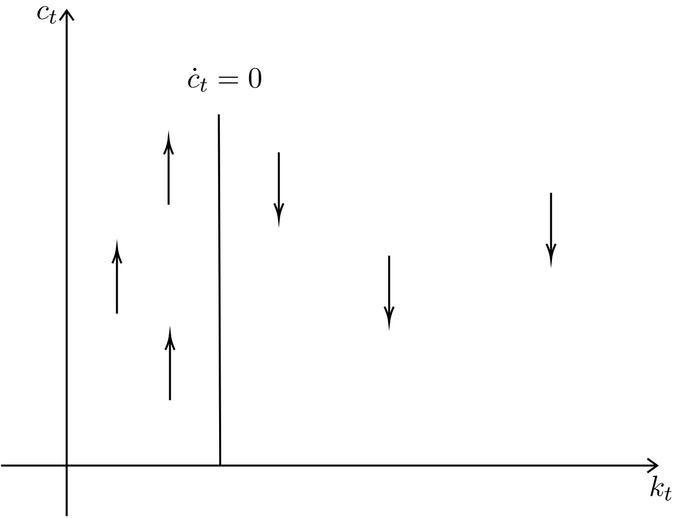
Transition Dynamics

- Remember that the equilibrium is characterized by the equations (+ TVC and k_0):

$$\frac{\dot{c}_t}{c_t} = \frac{(f'(k_t) - \delta - \rho)}{\sigma(c_t)},$$
$$\dot{k}_t = f(k_t) - (\delta + n)k_t - c_t.$$

- How can we analyze the dynamics of the system outside the steady state? $\Rightarrow$ Phase diagram.
- We will also show that the system is *saddle-path stable*: there exists a unique trajectory $\{k_t, c_t\}$ that converges to the steady state.
  - Given the state k_0, the control c_0 (or alternatively μ_0) adjusts instantly to the unique trajectory. That's why the control variable is known as the ***jump variable***.
  - For example, if an unexpected policy change occurs, the consumer adjusts c_t to the optimal path immediately while the state k necessarily follows the LOM.

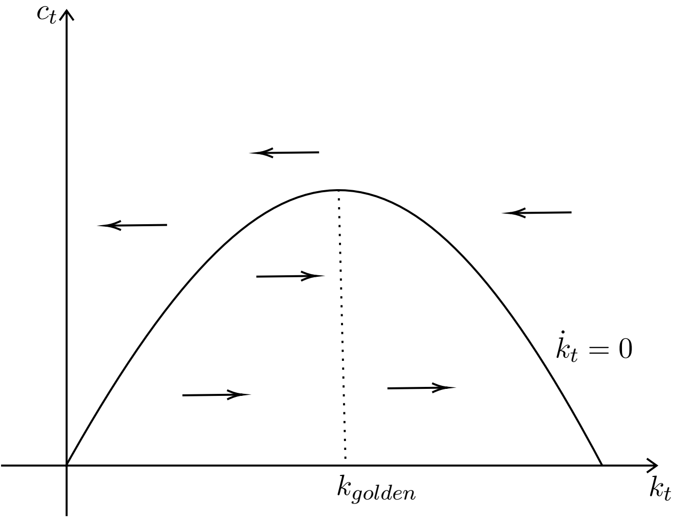
Phase Diagram



$$\frac{\dot{c}_t}{c_t} = \frac{(f'(k_t) - \delta - \rho)}{\sigma(c_t)}$$

- If $\uparrow k_t \Rightarrow \downarrow f'(k_t) \Rightarrow \downarrow \dot{c}$.
- The vertical line represents the space where consumption is constant.
- The line is vertical as depends only on capital (and not on consumption): $f'(k_t) = \delta + \rho$.

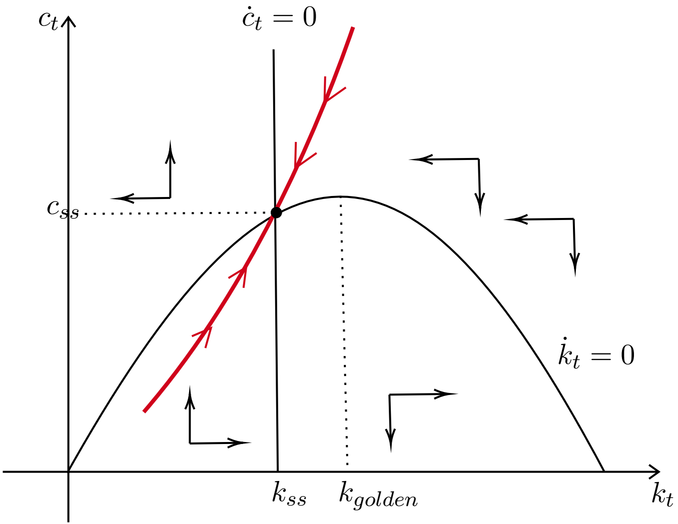
Phase Diagram



$$\dot{k}_t = f(k_t) - (\delta + n)k_t - c_t$$

- If $\uparrow c_t \Rightarrow \downarrow \dot{k}$.
- Capital is constant at the inverted U-shape line: $f(k_t) - (\delta + n)k_t$.
- k_{golden} is the point that maximizes consumption.

Phase Diagram



- Given k_0, c_0 "jumps" to the *stable-path*.
- If $c_0' > c_0$ the path converges to $k = 0$ and $c > 0$: violates the feasibility condition.
- If $c_0'' < c_0$ the path converges to $c = 0$ and $k > 0$: violates the TVC.
- There is only one path that converges to the steady state.

Local Stability

- Another way to check the *Saddle-path Stability* of the system is to look at the local stability conditions.
- Linearize the system equations using Taylor expansion around the Steady State:

$$\dot{k}_t = f(k_t) - (\delta + n)k_t - c_t \Rightarrow \dot{k} = (f'(k_{ss}) - (\delta + n))(k - k_{ss}) - (c - c_{ss})$$
$$\dot{c}_t = c_t \frac{(f'(k_t) - \delta - \rho)}{\sigma} \Rightarrow \dot{c} = c_{ss} \frac{f''(k_{ss})}{\sigma}(k - k_{ss}) + \underbrace{\frac{(f'(k_{ss}) - \delta - \rho)}{\sigma}}_{=0}(c - c_{ss})$$

- Write the system in the form:

$$\begin{bmatrix} \dot{k} \\ \dot{c} \end{bmatrix} = \begin{bmatrix} f'(k_{ss}) - \delta - n & -1 \\ c_{ss} f''(k_{ss})/\sigma & 0 \end{bmatrix} \begin{bmatrix} k - k_{ss} \\ c - c_{ss} \end{bmatrix}$$

Local Stability

Theorem (Acemoglu 7.19)

Consider the system $\dot{x}_t = G(x_t)$ where G is continuously differentiable and x_0 given. The steady state is $G(x^*) = 0$ and define $A = DG(x^*)$, D is the jacobian of G.
Suppose m eigenvalues of A have negative real parts while $n - m$ have positive real parts.
Then there exists a m-dimensional manifold (i.e., a topological space) in the neighborhood of the steady state, such that from any x_0 on that manifold, there is a unique $x_t \to x^*$.

- In our case if the matrix

$$A = \begin{bmatrix} f'(k_{ss}) - \delta - n & -1 \\ c^* f''(k_{ss})/\sigma & 0 \end{bmatrix}$$

has $m = 1$ negative eigenvalues, then there exists a line (manifold dimension $m = 1$) of points (c, k) that converges to the steady state.

Local Stability

- We want to find the eigenvalues λ such that $Ax = \lambda x$. In this case we have $\det(A - \lambda I)x = 0$

$$\det \begin{bmatrix} f'(k_{ss}) - \delta - n - \lambda & -1 \\ c^* f''(k_{ss})/\sigma & 0 - \lambda \end{bmatrix} = 0$$

$$\det(A - \lambda I) = -\lambda[f'(k_{ss}) - \delta - n - \lambda] + c_{ss} f''(k_{ss})/\sigma = 0$$

so

$$\lambda = [f'(k_{ss}) - \delta - n \pm \sqrt{(f'(k_{ss}) - \delta - n)^2 - 4c_{ss} \underbrace{f''(k_{ss})}_{<0} /\sigma}]/2$$

- Since $\sqrt{(f'(k_{ss}) - \delta - n)^2 - 4c_{ss} f''(k_{ss})/\sigma} > f'(k_{ss}) - \delta - n$, there exists exactly one negative eigenvalue.

Balanced Growth Path

- For the model to be consistent with the Facts of Kaldor it has to have long-run growth.
- **Balanced Growth Path**: All variables grow at a constant rate.
- Suppose a technology with *Labor-augmenting Technological Change*:

$$Y_t = F(K_t, A_t L_t), \quad \text{where } A_t = A_0 e^{gt}$$

  and g is the growth rate.
- We redefine the variables in labor efficient units so that they remain stationary: $\tilde{y}_t = y_t / A_t$, $\tilde{k}_t = k_t / A_t$, $\tilde{c}_t = c_t / A_t$.

Balanced Growth Path

- Note that the growth of the new variables is the per capita growth rate minus the technological advance:

$$\frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{\dot{c}}{c} - \frac{\dot{A}_t}{A_t} = \frac{\dot{c}}{c} - g \quad \text{and} \quad \frac{\dot{\tilde{k}}}{\tilde{k}} = \frac{\dot{k}}{k} - g$$

- Using this argument, the fact that F is CRS ($F(k, A) = AF(\tilde{k}, 1)$), and the definition of $\dot{k}$:

$$\begin{aligned} \frac{\dot{\tilde{k}}}{\tilde{k}} &= \frac{F(\tilde{k}_t, 1)A_t - (\delta + n)\tilde{k}_t A_t - \tilde{c}_t A_t}{k_t} - g \\ \dot{\tilde{k}} &= f(\tilde{k}_t) - (\delta + n + g)\tilde{k}_t - \tilde{c}_t \end{aligned}$$

Balanced Growth Path

- And the stationary Euler Equation:

$$\frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{(F_k(k_t, A_t) - \delta - \rho)}{\sigma(c_t)} - g$$
$$= \frac{(f'(\tilde{k}_t) - \delta - \rho)}{\sigma(c_t)} - g,$$

where we used the fact that

$$\frac{\partial F(k,A)}{\partial k} = \frac{\partial F(A\tilde{k}, A)}{\partial \tilde{k}} \frac{\partial \tilde{k}}{\partial k} = A f'(\tilde{k}) \frac{1}{A} = f'(\tilde{k}).$$

- By a similar argument we have that $r_{ss} = f'(\tilde{k})$ is constant in the long run (Kaldor's Facts), i.e., $r_t \to r_{ss}$.

Balanced Growth Path

- The only way for $\dot{\tilde{c}} = 0$, is if per capita consumption grows at a constant rate in the long run: $\dot{c}_t / c_t \to g$.
- By the Euler Equation, this implies that $\sigma(c_t) \to \sigma$.
- **Condition for BGP** is that the elasticity of marginal utility of consumption is asymptotically constant. Or alternatively, that the intertemporal elasticity of substitution is asymptotically constant.
- That's why the CRRA utility is so commonly used:

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}$$

Balanced Growth Path

- Given that $\sigma(c_t) \to \sigma$ is constant in the long run. The condition for utility to be bounded now is: $\rho - n > g(1-\sigma)$. Intuition:

$$\int_0^\infty e^{-(\rho-n)t} \frac{c_t^{1-\sigma}}{1-\sigma} dt$$

$$\int_0^\infty e^{-(\rho-n)t} \frac{(\tilde{c}_t A_0 e^{gt})^{1-\sigma}}{1-\sigma} dt$$

$$\int_0^\infty e^{-(\rho-n-g(1-\sigma))t} \frac{(\tilde{c}_t A_0)^{1-\sigma}}{1-\sigma} dt$$

Balanced Growth Path

- The solution now is the system of equations:

$$\frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{(f'(\tilde{k}_t) - \delta - \rho - g\sigma)}{\sigma}$$
$$\dot{\tilde{k}}_t = f(\tilde{k}_t) - (\delta + n + g)\tilde{k}_t - \tilde{c}_t$$

- Along with the initial condition $\tilde{k}_0$ and the TVC:

$$\lim_{t\to\infty} e^{-(\rho-n-g(1-\sigma))t} u'(\tilde{c}_t)\tilde{k}_t = 0$$

Steady State

- Note that now the steady-state capital depends on the form of utility (σ):

$$f'(\tilde{k}_{ss}) = \delta + \rho + g\sigma,$$

  which implies that: $r_{ss} = \delta + \rho + g\sigma$!

- $\uparrow \sigma \rightarrow$ lower intertemporal elasticity of substitution $\rightarrow \downarrow \tilde{k}_{ss}$.
- In a way, very similar to Solow:
  - $\tilde{k}$ is endogenous and depends on δ, g, and discount/IES determines savings (Solow: exogenous savings rate, but depends on n).
  - Long-run per capita growth is exogenous and given by g (just like in Solow).

Example

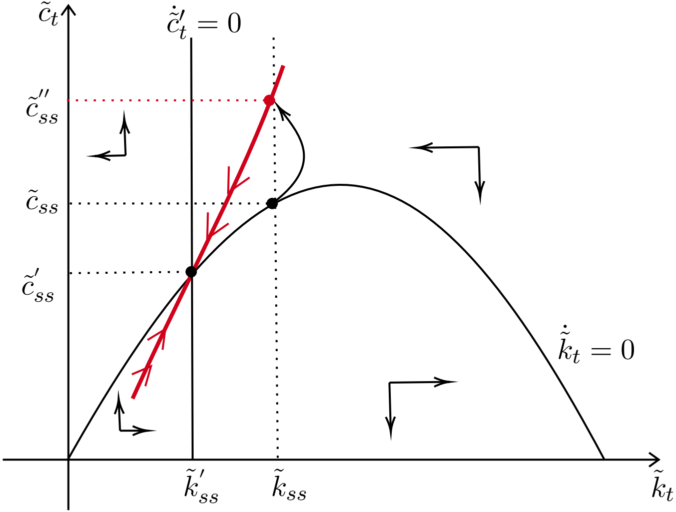
- Consider CRRA utility and Cobb-Douglas production function, $Y_t = F(K_t, A_t L_t) = K_t^\alpha (A_t L_t)^{1-\alpha}$:
  - $\tilde{y}_t = \tilde{k}^\alpha$, where for an arbitrary aggregate variable X, $\tilde{x} = X/(AL)$.
  - $r = f'(\tilde{k}) = \alpha \tilde{k}^{\alpha-1}$.
- The Euler Equation and the resource constraint:

$$\begin{aligned}
\frac{\dot{\tilde{c}}}{\tilde{c}} &= \frac{(r_t - \delta - \rho)}{\sigma(c_t)} - g = \frac{1}{\sigma}(\alpha \tilde{k}^{\alpha-1} - \delta - \rho - \sigma g) \\
\dot{\tilde{k}} &= f(\tilde{k}_t) - (\delta + n + g)\tilde{k}_t - \tilde{c}_t = \tilde{k}^\alpha - (\delta + n + g)\tilde{k}_t - \tilde{c}_t
\end{aligned}$$

- Steady state:

$$\tilde{k}_{ss} = \left(\frac{\alpha}{\delta + \rho + \sigma g}\right)^{1/(1-\alpha)} \quad \text{and} \quad \tilde{c}_{ss} = \left(\frac{\alpha}{\delta + \rho + \sigma g}\right)^{\frac{1}{1-\alpha}} \left(\frac{(\delta + \rho + \sigma g) - \alpha(\delta + n + g}{\alpha}\right.$$

Comparative Dynamics: Increase in ρ

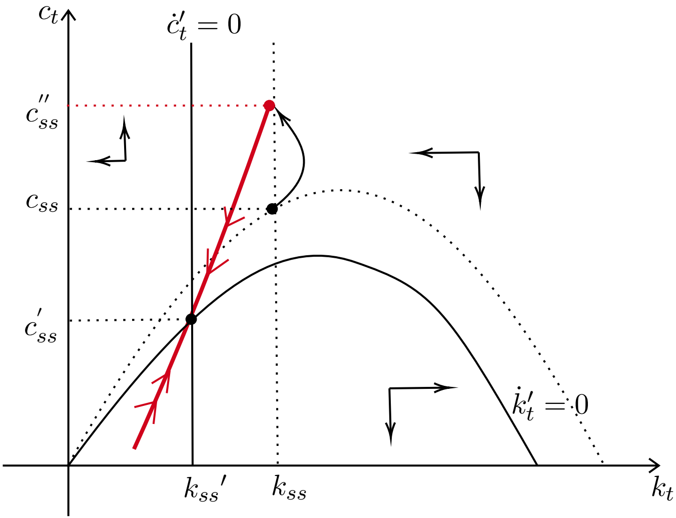


- Suppose the economy is in the SS and ρ increases.

$$\frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{(f'(\tilde{k}_t) - \delta - \rho - g\sigma)}{\sigma}$$

- The line $\dot{\tilde{c}}$ shifts to the left and $\tilde{c}$ jumps to the new *stable-path*.
- Eventually the system converges to the new SS.

Comparative Dynamics: Increase in δ



- Suppose the economy is in the SS and δ increases.

$$\frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{(f'(\tilde{k}_t) - \delta - \rho - g\sigma)}{\sigma}$$

$$\dot{\tilde{k}} = f(\tilde{k}_t) - (\delta + n + g)\tilde{k}_t - \tilde{c}_t$$

- The line $\dot{\tilde{c}}$ shifts to the left and the line $\dot{\tilde{k}}$ shifts downward.
- $\tilde{c}$ jumps, and eventually the system converges to the new SS.

Solving for the Equilibrium Path Numerically

- We saw that the Neoclassical Growth Model cannot be solved analytically (except in very special cases). This means, we must solve it numerically.
- The simplest method to solve the transition dynamics is using **shooting algorithm**.
- The main idea:
  - We have the initial condition k_0 (or the initial steady state) and the final steady state (k_{ss} and c_{ss}).
  - We can use the solution system to connect the initial to the terminal value. In discrete time, the system is:

$$u'(c_t) = (f'(k_{t+1}) + 1 - \delta)\beta u'(c_{t+1}) \qquad \text{(Euler Equation)}$$

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t \qquad \text{(Resource Constraint)}$$

Shooting Algorithm

- **Shooting algorithm:** We guess an initial value of c_0 and use the system of equations to simulate the sequence $\{k_{t+1}, c_t\}_{t=0}^{S}$ until a period S.
  - If we stop at the final steady state, c_0 is the solution;
  - otherwise, we try another c_0.
- It is an intuitive algorithm that works to solve systems of difference (or differential) equations with two conditions. It works in discrete and continuous time.
- Recall the phase diagram. We are trying to guess the initial jump of the control variable.
- If we miss the initial jump of c_0, the path diverges from the steady state.
- We will use it more later when we study fiscal policy.

Shooting Algorithm

(i) Solve for the final steady state: k_{ss} and c_{ss}.

(ii) Select a sufficiently long time period S (so that the economy reaches the SS), and guess an initial consumption solution candidate c_0.

(iii) Use the *Resource Constraint* and k_0 to compute k_1. Use c_0 and k_1 in the Euler Equation to compute c_1. Continue using c_t and k_t to find c_{t+1} and k_{t+1}.

(iv) Proceed until period S to find the candidate solution sequence: $\{\hat{k}_{t+1}, \hat{c}_t\}_{t=0}^{S}$.

(v) Compute $\hat{k}_S - k_{ss}$. If $\hat{k}_S > k_{ss}$, increase the guess c_0 and try again; If $\hat{k}_S < k_{ss}$, decrease the guess c_0 and try again.

(vi) Proceed until finding a c_0 such that $\hat{k}_S \approx k_{ss}$.

Taking Stock

- Neoclassical growth model: explains the convergence process among different countries.
  - Many conclusions are similar to the Solow model.
- Endogenous savings bring new insights into the impact of preferences, taxation, etc., on long-term growth.
- For the Balanced-Growth Path, we need preferences with constant elasticity of substitution.
- Does not explain very long-term growth $\Rightarrow$ grows at the exogenous rate g.
  - Motivation to develop endogenous growth models.

Application: Real Business Cycles

What Are Business Cycles?

- Business cycles are the **short-run fluctuations** of economic activity around its long-run trend.

- One of the earliest questions in macroeconomics:
  - Why do economic cycles exist?
  - How do they work (what are the propagation mechanisms and shocks)?
  - Can we do anything about it?

- The basic Real Business Cycles model by **Kydland and Prescott (1982)**: stochastic neoclassical growth model with endogenous labor supply and TFP shocks.

- Merged long-run growth and short-run cycles in one framework.

A Brief History of Business Cycle Theory

- **1950s–1970s**: Keynesian simultaneous-equation models dominated.
  - Failed to explain the stagflation of the 1970s (supply-side shocks).
  - The **Lucas Critique**: reduced-form relationships are not policy-invariant. Models must be built on micro-foundations with rational expectations.
- **Lucas and Rapping (1969)**: introduced rational expectations and *intertemporal labor substitution* as drivers of employment fluctuations.
- Subsequent attempts used monetary (unanticipated) shocks but had limited quantitative success.
- **Kydland and Prescott (1982)**: after discarding monetary shocks, found that *technology shocks alone* could replicate $\approx 70\%$ of U.S. business cycle fluctuations.
  - Internal consistency + quantitative success, Kydland and Prescott's (1982) RBC became the great successor of the Rational Expectations Revolution.
  - Launched DSGE modeling and the *calibration + simulation* methodology.

Initial success of RBC

"As state by Plosser, that such a simple model "with no government, no money, no market failures of any kind, rational expectations, no adjustment costs and identical agents could replicate actual experiences this well is most surprising". What made the Kydland and Prescott model stunning was that, while resting on just one shock and six parameters it delivered as much as models containing dozens of equations and many more free parameters."
- De Vroey (2015, p. 266)

Isolating the Cycle: HP Filter

- Decompose a time series (in logs) into trend and cyclical component: $Y_t = \tau_t + Y_{c,t}$.
- The **Hodrick-Prescott (HP) filter** extracts τ_t by solving:

$$\min_{\{\tau_t\}} \sum_{t=1}^{T} (Y_t - \tau_t)^2 \; + \; \lambda \sum_{t=2}^{T-1} \left[(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1}) \right]^2$$

- λ controls smoothness: standard choice $\lambda = 1{,}600$ for quarterly data.
  - $\lambda \to 0$: trend tracks data perfectly. $\lambda \to \infty$: linear trend.
- Many criticisms and alternatives to the HP filter (band-pass, first differences). See Stock and Watson (1999, Handbook of Macroeconomics) and Hamilton (2018, ReStats).

HP Filter

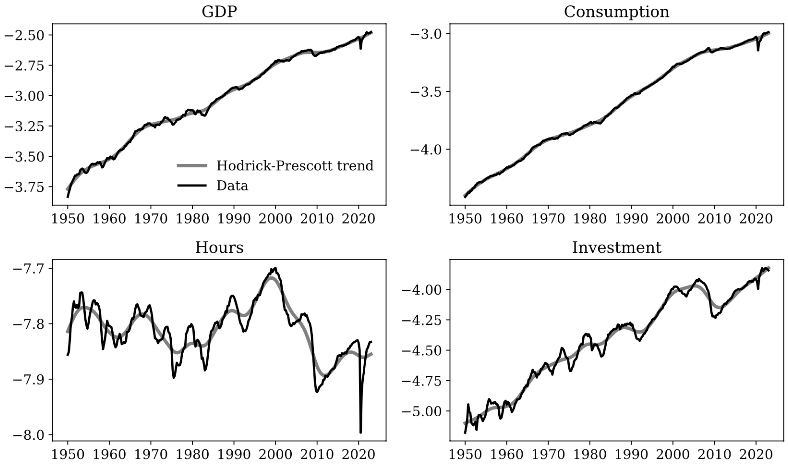


Figure 14.4: Actual and trends of logs of U.S. aggregates

Source: PhD Macrobook Ch. 14.

Stylized Business Cycle Facts

HP-filtered quarterly U.S. data (1949–2022). **Source**: PhD Macrobook (2025, Ch. 14):

Variable	σ_x (%)	Rel. σ_x/σ_y	Auto-corr	$\rho(x_t, y_t)$
Output y	1.63	1.00	0.78	1.00
Consumption c	1.07	0.65	0.64	0.75
Gov. Consumption g	3.02	1.85	0.89	0.15
Investment i	4.37	2.68	0.86	0.77
Employment	1.58	0.97	0.81	0.79
Hours ℓ	2.11	1.29	0.80	0.86
Unemployment	15.63	9.57	0.80	-0.83
Lab. Productivity	1.15	0.70	0.72	0.27
Wages	1.24	0.76	0.72	-0.00
Price Level	0.92	0.56	0.92	-0.10
TFP z	0.90	0.55	0.75	0.51

All variables in logs and HP-filtered ($\lambda = 1{,}600$), except unemployment rate and Fed funds rate.

Stylized Business Cycle Facts

1. **Consumption smoothing**: Consumption is less volatile than output ($\sigma_c/\sigma_y \approx 0.65$). Durable goods are more volatile.
2. **High investment volatility**: Investment is the most volatile component ($\sigma_i/\sigma_y \approx 2.7$).
3. **Labor market**: Hours are roughly as volatile as output. Employment is slightly less volatile; unemployment is highly volatile and *countercyclical*.
4. **Cyclicality**: Consumption, investment, employment, hours, and TFP are all procyclical. Government consumption is roughly acyclical; unemployment is strongly countercyclical.
5. **Price and wage dynamics**: Despite large swings in quantities, real wages and the price level are much less volatile and essentially acyclical.
6. **Persistence**: All major aggregates are highly serially correlated ($\rho \approx 0.8$). TFP and labor productivity are about as persistent as output itself.

Business Cycle Facts: Emerging Markets

TABLE 1
EMERGING VS. DEVELOPED MARKETS (Averages)

	Emerging Markets	Developed Markets
$\sigma(Y)$	$\boxed{2.74}$ (.12)	1.34 (.05)
$\sigma(\Delta Y)$	1.87 (.09)	.95 (.04)
$\rho(Y)$	.76 (.02)	.75 (.03)
$\rho(\Delta Y)$	.23 (.04)	.09 (.03)
$\sigma(C)/\sigma(Y)$	$\boxed{1.45}$ (.02)	.94 (.04)
$\sigma(I)/\sigma(Y)$	3.91 (.01)	3.41 (.01)
$\sigma(TB/Y)$	$\boxed{3.22}$ (.17)	1.02 (.03)
$\rho(TB/Y, Y)$	−.51 (.04)	−.17 (.04)
$\rho(C, Y)$	.72 (.04)	.66 (.04)
$\rho(I, Y)$	.77 (.04)	.67 (.04)

NOTE.—This table lists average values of the moments for the group of emerging (13) and developed (13) economies. The values for each country separately are reported in table 2. Data are Hodrick-Prescott filtered using a smoothing parameter of 1,600. The standard deviations are in percentages. The standard errors for the averages were computed assuming independence across countries. The definition of an emerging market follows the classification in Standard & Poor's (2000).

Source: Aguiar and Gopinath (2007).

Conditional Data Moments

- Stylized facts above are *unconditional* moments: they tell us *what* co-moves but not *why*.
- To trace the role of specific shocks, we use **impulse response functions (IRFs)**.
- Identification is not trivial. Many different methods (SVARs, Local projecions).
- The book uses a structural VAR with timing restrictions (TFP does not respond contemporaneously to output or investment) to identify TFP shocks:
  - Positive co-movement of TFP, output, and investment after the shock.
  - Persistent responses consistent with the propagation mechanism below.
- **Key use**: compare the model's theoretical IRFs to those estimated from data — the standard evaluation metric in modern business cycle analysis.

Conditional Data Moments

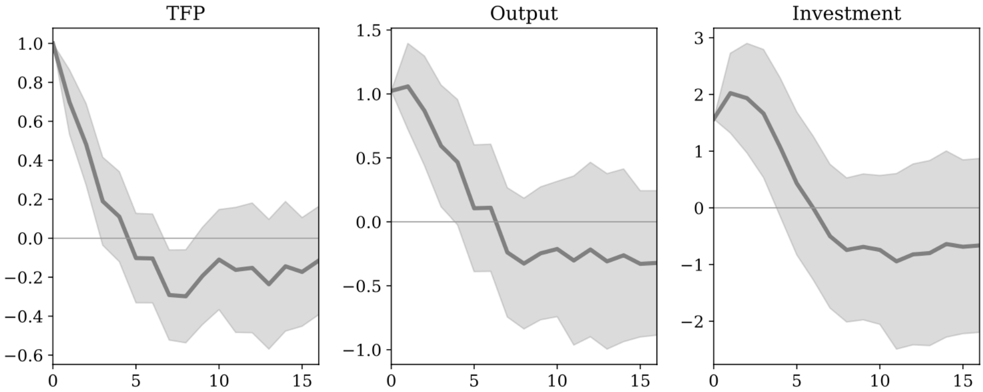


Figure 14.6: Impulse responses to TFP shock estimated with structural VAR

Source: PhD Macrobook (2025, Ch. 14)

The Core RBC Model

Core RBC Model: Environment

- The RBC model is a **stochastic neoclassical growth model** with elastic labor supply.
- **Environment**:
  - Discrete time, representative household, infinite horizon.
  - No population or productivity trend growth.
  - Competitive markets, no government.
  - Welfare Theorems hold $\Rightarrow$ social planner's problem $=$ competitive equilibrium.
- Two key departures from the deterministic NGM:
  1. **Elastic labor supply**: households choose how much to work each period.
  2. **Stochastic TFP**: z_t follows an AR(1) process.

Preferences

- The representative household values consumption c_t and *leisure* $1-\ell_t$, where $\ell_t \in [0,1]$ is the fraction of time allocated to work:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t, 1-\ell_t), \qquad \beta \in (0,1)$$

  where U is increasing and concave in both arguments and satisfies Inada conditions.

- Budget constraint (household owns capital and rents it to firms):

$$c_t + k_{t+1} \leq (1 + r_t - \delta)\, k_t + w_t \ell_t \qquad \forall\, t$$

  along with a no-Ponzi condition and $k_0 > 0$ given.

- r_t is the rental rate of capital and w_t is the real wage, both taken as given by the household.

Technology

- A representative firm produces output using capital k_t and labor ℓ_t:

$$y_t = z_t F(k_t, \ell_t) = z_t k_t^{\alpha} \ell_t^{1-\alpha}, \qquad \alpha \in (0,1)$$

  CRS $\Rightarrow$ zero profits in equilibrium. Capital accumulates as $k_{t+1} = (1-\delta)k_t + i_t$.

- Profit maximization yields the **input demand equations**:

$$r_t = z_t F_k(k_t, \ell_t) = \alpha\, z_t \left(\frac{k_t}{\ell_t}\right)^{\alpha-1}$$

$$w_t = z_t F_\ell(k_t, \ell_t) = (1-\alpha)\, z_t \left(\frac{k_t}{\ell_t}\right)^{\alpha}$$

- Stochastic TFP z_t follows an AR(1) in logs:

$$\log z_t = \rho_z \log z_{t-1} + \sigma_z \varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} N(0,1)$$

  where $\rho_z \in (0,1)$ governs persistence and $\sigma_z > 0$ controls the size of innovations.

Equilibrium Conditions

- Equilibrium requires that at every t:
  - **Goods market**: $y_t = c_t + i_t \qquad \forall\, t$
  - **Capital market** (supply = demand): $a_t^s = k_t^d \qquad \forall\, t$
  - **Labor market** (supply = demand): $\ell_t^s = \ell_t^d \qquad \forall\, t$
- Prices $(r_t,\, w_t)$ are those that clear both factor markets.

Household's Problem

- Substituting $1-\ell_t$ for leisure, the Lagrangian is:

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t, 1-\ell_t) + \lambda_t[(1+r_t-\delta)\, a_t + w_t\ell_t - c_t - a_{t+1}]$$

- First-order conditions:
  - c_t: $\quad \beta^t U_1(c_t, 1-\ell_t) = \lambda_t$
  - ℓ_t: $\quad \beta^t U_2(c_t, 1-\ell_t) = \lambda_t w_t \qquad$ (note: $U_2 < 0$, $w_t > 0$, $U_2 = -U_{1-\ell}$)
  - a_{t+1}: $\quad \lambda_t = \mathbb{E}_t[(1+r_{t+1}-\delta)\,\lambda_{t+1}]$
- Combining the FOCs gives two key optimality conditions (at every t):

$$U_1(c_t, 1-\ell_t) = \beta\,\mathbb{E}_t[(1+r_{t+1}-\delta)\,U_1(c_{t+1}, 1-\ell_{t+1})] \tag{EE}$$

$$\frac{U_2(c_t, 1-\ell_t)}{U_1(c_t, 1-\ell_t)} = w_t \tag{LS}$$

- (EE): standard intertemporal Euler equation. (LS): MRS $=$ wage (intratemporal).

Decentralized Equilibrium

- The decentralized equilibrium is characterized by the following system for all t:

$$
\begin{aligned}
&U_1(c_t, 1-\ell_t) = \beta\,\mathbb{E}_t[(1 + r_{t+1} - \delta)\,U_1(c_{t+1}, 1-\ell_{t+1})] \\
&U_2(c_t, 1-\ell_t) = U_1(c_t, 1-\ell_t)\,w_t \\
&r_t = z_t F_k(k_t, \ell_t), \quad w_t = z_t F_\ell(k_t, \ell_t) \\
&y_t = c_t + i_t, \quad k_{t+1} = (1-\delta)k_t + i_t \\
&y_t = z_t F(k_t, \ell_t), \quad \log z_t = \rho_z \log z_{t-1} + \sigma_z \varepsilon_t
\end{aligned}
$$

  along with the TVC and k_0 given.

- Depending on the functional forms the system can be reduced further.
- **First and Second Welfare Theorems hold**: the competitive equilibrium is Pareto efficient, so one can solve the *social planner's problem* to characterize all allocations.

Functional Forms: KPR Preferences

- For the model to be consistent with **balanced growth** (constant hours on the BGP), King–Plosser–Rebelo (1988) show preferences must take the form:

$$U(c, 1-\ell) = \frac{\left(c\, v(1-\ell)\right)^{1-\sigma} - 1}{1-\sigma}$$

where v is strictly increasing and concave with $v'(0) = \infty$.

- Common specification: Cobb-Douglas aggregator, $\sigma = 1$ (log utility):

$$u(c, 1-\ell) = (1-\phi)\log c + \phi \log(1-\ell), \qquad \phi \in (0,1)$$

- The labor supply condition (LS) then becomes:

$$\frac{\phi}{1-\phi} \cdot \frac{c_t}{1-\ell_t} = w_t$$

- Intertemporal elasticity of substitution $= 1/\sigma$; ϕ governs the elasticity of labor supply.

Functional Forms: Other Preferences

- In practice, many papers use preferences **not consistent with BGP**. For example, the standard CRRA utility with separable disutility of labor:

$$u(c, 1-\ell) = \frac{c^{1-\sigma} - 1}{1-\sigma} - \theta \, \frac{\ell^{1+\eta}}{1+\eta}$$

where $1/\sigma$ is the IES and $1/\eta$ is the Frisch elasticity. Labor supply: $\theta \, \ell_t^{\eta} = c_t^{-\sigma} w_t$.

- **Greenwood–Hercowitz–Huffman (GHH) preferences**:

$$u(c, 1-\ell) = \frac{(c + v(1-\ell))^{1-\sigma} - 1}{1-\sigma}$$

- GHH are widely used when we want to **eliminate the wealth effect on labor supply**.
  - Labor supply: $\theta \, \ell_t^{\eta} = w_t$ — depends only on the wage, not on consumption.
  - Eliminates the income effect on labor supply.

Shocks, Labor Supply, and Propagation

Labor-Leisure Decision

- Before studying the impact of shocks, it is important to understand how agents respond to a wage change.

- The labor supply condition (LS) with KPR log preferences:

$$\frac{\phi}{1-\phi} \cdot \frac{c_t}{1-\ell_t} = w_t$$

- Suppose for a moment that c_t is held constant. An increase in w_t raises ℓ_t: this is the **substitution effect** (leisure becomes more expensive).

- A wage increase also makes households wealthier: they want to consume more goods *and* more leisure (work less): this is the **income effect**.

- For realistic calibrations, the **substitution effect dominates** the income effect $\Rightarrow$ labor supply is upward sloping in wages.

Frisch Elasticity

- The **Frisch elasticity** measures the response of labor supply to a wage change *holding the marginal utility of wealth constant*:

$$\varepsilon_F \equiv \left.\frac{d\log \ell_t}{d\log w_t}\right|_{\lambda_t=\bar{\lambda}}$$

- With KPR log preferences: $\varepsilon_F = \dfrac{1-\ell}{\ell}$.

→ Derivation

  - At $\ell^* = 1/3$: $\varepsilon_F = 2$. High, but needed for quantitative success.

- **Micro–macro tension**:
  - Microeconomic estimates: $\varepsilon_F \approx 0$–0.5 (intensive margin).
  - Model requires $\varepsilon_F \approx 2$ to generate enough hours volatility.
  - Resolution: the *extensive margin* (employment) accounts for most aggregate hours variation — motivates indivisible labor models.

Intertemporal Labor Substitution

- Combining (EE) and (LS) with KPR log preferences (abstracting from uncertainty) yields an **intertemporal labor supply** condition:

$$\frac{1-\ell_{t+1}}{1-\ell_t} = \beta\left(1 + r_{t+1} - \delta\right)\frac{w_t}{w_{t+1}}$$

- Agents work more when wages are temporarily high relative to future wages.
- Agents work more when the real interest rate is high (higher return to saving $\Rightarrow$ work today, consume tomorrow).
- This **intertemporal labor substitution** is the key transmission channel of TFP shocks in the RBC model.

Transitory TFP Shock ($\rho_z \approx 0$)

- After a positive shock ($\uparrow z_t$): capital is *predetermined*, so ℓ_t and c_t jump immediately; $\uparrow w_t$ and $\uparrow r_t$ on impact.
- Since the shock is transitory, z_t returns to its mean next period.
- **Small wealth effect**: the shock has very little impact on permanent income.
  - Income effect on consumption is small $\Rightarrow$ c_t increases only a little.
  - Income effect on leisure is small $\Rightarrow$ **substitution effect dominates**: agents work much more today.
- High intertemporal labor substitution generates **large amplification** of output at t.
- However, since $z_{t+1} \approx \bar{z}$, output falls back quickly $\Rightarrow$ **little persistence**. The model cannot generate internal propagation.

TFP Shocks

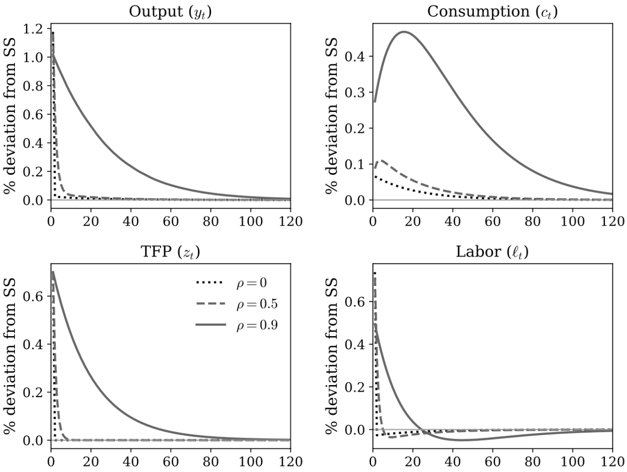


Figure 14.9: Sensitivity to TFP shock persistence in the RBC model

Source: PhD Macrobook (2025, Ch. 14)

Permanent TFP Shock ($\rho_z \approx 1$)

- The shock permanently raises $z_t \Rightarrow$ new steady state with higher k^*, y^*, c^* (as in the NGM).
- **Large wealth effect**: permanent income rises substantially.
  - Income effect on consumption is large $\Rightarrow c_t$ jumps up significantly.
  - Income effect on leisure is large $\Rightarrow$ **income and substitution effects offset each other**: labor response is muted.

$$\frac{\phi}{1-\phi} \cdot \frac{\uparrow c_t}{1-\ell_t} = \uparrow w_t$$

- **Less amplification** in labor and output on impact compared to the transitory case.
- **More persistence**: dynamics follow the transition path to the new steady state, driven by capital accumulation — just like the deterministic NGM.

Quantitative Performance

Solution Methods

- The RBC equilibrium is a system of nonlinear stochastic difference equations $\Rightarrow$ no closed-form analytical solution in general.
- Two main approaches:
  - **Dynamic Programming (global methods)**: solve the Bellman equation on a grid for (k, z). Exact nonlinear policy function solution.
  - **Perturbation methods (local methods)**: log-linearize the equilibrium conditions around the deterministic steady state; solve the resulting linear system.
- Most quantitative DSGE packages (`Dynare`, `SolveDSGE.jl`, `PyMacLab`, among others) use some form of log-linearization or some more general form of perturbation.
  - For more details on Perturbation, see Fernández-Villaverde, Rubio-Ramírez, & Schorfheide (2016, Handbook of Macro) and Schmitt-Grohé & Uribe (2004, JEDC).

Solution Methods: DP vs. Perturbation

- **Dynamic Programming**
  - Global Method (solution is a nonlinear policy function).
  - Slow (curse of dimensionality!).
  - Captures nonlinearities, asymmetries, etc.
  - Can be applied to non-convexities, discrete choice.
- ***Perturbation Methods***.
  - Local Method (what happens when the shock is very large? Covid?)
  - Fast.
  - Requires problem to be differentiable (possible, but complicated to deal with kinks).
  - Presents *Certainty Equivalence* (at first order). To capture uncertainty, risk, asymmetry, etc., higher-order approximations are needed.

Calibration

- Parameters are chosen to match **long-run facts** (Kaldor facts) and micro estimates:
  - $\alpha = 0.36$: capital share of income.
  - $\beta = 0.99$: quarterly discount factor (annual real rate $\approx 4\%$).
  - $\delta = 0.025$: quarterly depreciation ($\approx 10\%$ per year).
  - $\sigma = 1$ (log utility): consistent with BGP and micro evidence.
  - $\phi = 0.6325$: chosen so steady-state $\ell^* = 1/3$.
- Shock parameters estimated from the **Solow residual**:

$$\log z_t = \log y_t - \alpha \log k_t - (1-\alpha) \log \ell_t$$

Detrend with HP filter, then estimate AR(1): $\rho_z \approx 0.95$, $\sigma_z \approx 0.007$.

Quantitative Performance

	U.S. Data (1949–2022)			Core RBC Model		
Variable	Rel. σ	Auto-corr	$\rho(x, y)$	Rel. σ	Auto-corr	$\rho(x, y)$
Output y	1.00	0.78	1.00	1.00	0.73	1.00
Consumption c	0.65	0.64	0.75	0.32	0.81	0.90
Investment i	2.68	0.86	0.77	3.10	0.72	0.99
Hours ℓ	1.29	0.80	0.86	0.48	0.72	0.98
TFP z	0.55	0.75	0.51	0.69	0.43	0.99

Successes

Output, investment & consumption volatilities matched; procyclicality; high persistence.

Shortcomings

Hours too smooth; correlations too high (one shock); little internal propagation; wages too procyclical.

Extensions

Indivisible Labor (Hansen 1985, Rogerson 1988)

- **Problem**: Most hours adjustment in the data is along the *extensive margin* (employment). Standard model generates too little hours volatility even with $\varepsilon_F = 2$.

- **Solution**: Workers face a binary choice — work $\hat{\ell}$ hours or not work at all.
  - Discrete choice $\Rightarrow$ non-convexities. Resolved by employment lotteries (complete markets).

- **Result**: Aggregating over lotteries, the representative household's utility becomes:

$$U(c_t, \ell_t) = \log c_t - B\ell_t$$

  Labor is *linear* in utility $\Rightarrow$ aggregate Frisch elasticity $= \infty$.

- The model generates much greater hours volatility and reconciles macro aggregates with low intensive-margin micro Frisch elasticity estimates.

Capital Adjustment Costs

- **Problem**: Investment is too volatile and the real interest rate is too procyclical.
- **Idea** (Hayashi 1982): firms incur real costs when changing the capital stock:

$$k_{t+1} = (1-\delta)k_t + i_t - \Psi(i_t, k_t), \qquad \Psi(i_t, k_t) = \frac{\psi}{2}\left(\frac{i_t}{k_t} - \delta\right)^2 k_t$$

- This introduces **Tobin's q**: shadow price of installed capital $q_t = 1 + \psi(i_t/k_t - \delta)$.
  - Invest when $q_t > 1$; disinvest when $q_t < 1$.
- Adjustment costs smooth investment responses over time and dampen the interest rate co-movement with output.

Variable Capacity Utilization

- **Problem**: Capital is fully predetermined $\Rightarrow$ limited amplification on impact.
- **Idea** (Greenwood, Hercowitz, Huffman 1988): firms choose the *utilization rate* $u_t \in [0, 1]$:

$$y_t = z_t (u_t k_t)^\alpha \ell_t^{1-\alpha}$$

- Higher utilization raises depreciation: $k_{t+1} = i_t + \big(1 - \delta(u_t)\big) k_t$, $\delta' > 0$, $\delta'' > 0$.
- Optimal utilization equates the return to capital with the marginal depreciation cost:

$$r_t = \delta'(u_t)$$

- A positive TFP shock $\Rightarrow \uparrow r_t \Rightarrow \uparrow u_t \Rightarrow$ amplifies effective capital services without requiring new investment. Requires adjusted measurement of the Solow residual.

Additional Shocks

- With a single shock, all variables are nearly perfectly correlated — at odds with the data.
- **Investment-specific technology (IST) shocks** (Greenwood, Hercowitz, Krusell 2000):

$$k_{t+1} = (1-\delta)k_t + q_t\, i_t$$

  Shocks to investment efficiency q_t affect the productivity of new capital. IST shocks account for $\approx 30\%$ of business cycle fluctuations and break the strong wage–hours correlation.

- **Government spending shocks**: $c_t + i_t + g_t = y_t$.
  - Acts as a resource drain $\Rightarrow$ negative income effect raises labor supply.
  - Helps reduce the high cross-correlations and introduces a countercyclical margin.

Business Cycle Accounting

- **Chari, Kehoe, McGrattan (2007)**: introduce four *wedges* as distortions to the core RBC equilibrium:
    1. **Efficiency wedge** z_t: enters the production function as TFP.
    2. **Labor wedge** $(1 - \tau_t^{\ell})$: distorts the labor–leisure condition (search frictions, taxes, market power).
    3. **Investment wedge** $(1 + \tau_t^{i})^{-1}$: distorts the capital Euler equation (financial frictions, capital income taxes).
    4. **Government spending wedge** g_t: resource drain from private use.
- **Finding**: efficiency and labor wedges explain $\approx 95\%$ of output fluctuations; the labor wedge is quantitatively the most important.
- Motivates models with explicit *labor market frictions* (search and matching, endogenous unemployment) as the key frontier beyond the basic RBC model.

Taking Stock: RBC and the Research Frontier

- The RBC model is a **stochastic neoclassical growth model**: it inherits all the tools developed in this course and adds elastic labor supply and aggregate uncertainty.

- Despite its simplicity, it generates quantitatively realistic patterns for output, investment, and consumption volatility and cyclicality.

- Its shortcomings motivate the main extensions in modern macro:
  - Nominal rigidities $\Rightarrow$ **New Keynesian (DSGE) models**.
  - Labor market frictions $\Rightarrow$ **Search and matching models**.
  - Household heterogeneity $\Rightarrow$ **Incomplete markets / HANK models**.

- The calibration–simulation–moment-matching methodology introduced by Kydland and Prescott remains the **core toolkit of modern quantitative macroeconomics**.

Appendix

Digression: Representative Agent

← Back

- Suppose that instead of a representative household, there is a continuum of households h represented by the interval $[0, 1]$.
  - Advantage of using unit population: aggregate value = average.
  - In general, $u()$ and β can depend on family h
  - Households can also be heterogeneous in their endowments: income, wealth...
- We are interested in studying aggregate variables $\Rightarrow$ eventually we have to aggregate the decisions of all individuals in the economy.
- That is, the aggregate household demand, C_t, is defined as the sum of all hh in the economy:

$$C_t = \int_0^1 c_t^h dh \tag{35}$$

where c_t^h is the optimal consumption of agent h.

Digression: Representative Agent

← Back

- **Problem:** Aggregating heterogeneous agents can be complicated.
- It implies solving the decision of each individual agent individually.
- Solution: Assume the existence of a representative agent.
  - The aggregate demand of the economy can be represented by a representative agent making decisions subject to the aggregate budget constraint.
  - When can we do this? What do we lose?
- **Trivial solution:** Assume that preferences and endowments are equal for all h:
  - $u^h() = u(), \quad \beta^h = \beta$ and equal endowments $\Rightarrow c^h = c$.
- We don't always need to assume that all agents are equal for our model to be represented by a representative household.

Gorman Aggregation Theorem

← Back

Theorem (Gorman Aggregation Theorem)

Consider an economy with $N < \infty$ goods and a set H of agents with wealth w^h. Suppose that the preferences of each household $h \in H$ are represented by the indirect utility

$$v^h(p, w^h) = a^h(p) + b(p)w^h, \tag{36}$$

then preferences can be aggregated and represented by an agent with indirect utility

$$v(p, w) = a(p) + b(p)w, \tag{37}$$

where $a(p) = \int_{h \in H} a^h(p)dh$ and $w = \int_{h \in H} w^h dh$.

- **Proof:** Use Roy's identity to find individual demand and take the integral over h.

Gorman Aggregation Theorem

← Back

- If preferences lead to linear indirect utilities in wealth with the same $b(p)$ for all agents, we can represent individual demand for any arbitrary good:

$$c^h(p, w^h) = \alpha^h(p) + \kappa(p) w^h \tag{38}$$

- **Linear** relationship between demand and wealth!
- **Intuition**:
  - If all agents have the same marginal propensity to consume, aggregate demand only depends on aggregate wealth!
  - When reallocating wealth from one agent to another, aggregate demand does not change.

A Simple Example

← Back

- Suppose 2 agents with Cobb-Douglas utility $U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$.
  - Capitalist receives profit $y^c = \pi$.
  - Worker receives wage $y^w = w$.
  - Aggregate income $Y = w + \pi$.
- Individual demands: $x_1^i = \alpha y^i / p_1$ and $x_2^i = (1-\alpha) y^i / p_2$ for $i = c, w$.
- Indirect utility:

$$v^i(p, y^i) = x_1^\alpha x_2^{1-\alpha} = \left(\alpha \frac{y^i}{p_1}\right)^\alpha \left((1-\alpha)\frac{y^i}{p_2}\right)^{1-\alpha} = \left(\frac{\alpha}{p_1}\right)^\alpha \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} y^i$$

- Indirect utility of the representative agent with income Y:

$$v(p, Y) = \left(\frac{\alpha}{p_1}\right)^\alpha \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} Y$$

Other Examples: Gorman Aggregation

← Back

(i) Quasi-homothetic utilities:

$$u(x_1^h, ..., x_N^h) = \left[\sum_{j=1}^{N}(x_j - \xi_j^h)^{(\sigma-1)/\sigma}\right]^{\sigma/(\sigma-1)} \tag{39}$$

define $\tilde{x}_j^h = x_j - \xi$. As long as the solution is interior, utility admits a representative agent with $\xi_j \equiv \int_h \xi_j^h dh$.

(ii) Quasi-linear utilities: $u(c, l) = u(c) + \phi l$.

Representative Agent

← Back

- There are versions of Gorman's Aggregation Theorem for dynamic economies.
- We assume u that allows for representative agents.
- From now on, we will represent households with a single representative household: $U(c_t^h) = U(c_t)$ (unless noted).
- **Exercise**: Find the representative agent of the economy with two agents from the previous section.

Example: Log Utility + Cobb-Douglas ($\delta = 1$)

← Back

- With $u(c) = \log(c)$, $\delta = 1$, $f(k) = k^\alpha$, the system becomes:

$$\frac{1}{c_t} = \frac{\alpha\beta k_{t+1}^{\alpha-1}}{c_{t+1}} \quad \text{(EE)}$$

$$c_t + k_{t+1} = k_t^\alpha \quad \text{(RC)}$$

- **Guess**: suppose the planner saves a constant fraction s of output in every period:

$$k_{t+1} = s\, k_t^\alpha, \qquad c_t = (1-s)\, k_t^\alpha.$$

- Substituting into the EE:

$$\frac{1}{(1-s)k_t^\alpha} = \frac{\alpha\beta\,(sk_t^\alpha)^{\alpha-1}}{(1-s)(sk_t^\alpha)^\alpha} = \frac{\alpha\beta}{(1-s)\, s\, k_t^\alpha}.$$

- Solving: $s = \alpha\beta$. **The guess is verified!**

Example: Log Utility + Cobb-Douglas ($\delta = 1$) — Solution

← Back

- The optimal policy functions are:

$$\begin{aligned} k_{t+1} &= \alpha\beta\, k_t^{\alpha} \\ c_t &= (1 - \alpha\beta)\, k_t^{\alpha} \end{aligned}$$

- The saving rate is $s = \alpha\beta$ — constant, regardless of k_t.
- **Intuition:** With log utility (elasticity of substitution = 1), income and substitution effects on saving cancel out exactly, yielding a constant saving rate.
- Note: this parallels the Solow model with $s = \alpha\beta$ — but here the saving rate is derived from optimization, not assumed.
- **Steady state**: $k_{ss}^{1-\alpha} = \alpha\beta \Rightarrow k_{ss} = (\alpha\beta)^{1/(1-\alpha)}$.

Discrete vs. Continuous Time: Euler Equation (CRRA)

← Back

- With CRRA utility $u(c) = c^{1-\sigma}/(1-\sigma)$, we have $u'(c) = c^{-\sigma}$.

- **Discrete time**: the Euler equation $u'(c_t) = \beta(1+r)u'(c_{t+1})$ becomes $c_t^{-\sigma} = \beta(1+r)c_{t+1}^{-\sigma}$. Taking logs:

$$\underbrace{\log c_{t+1} - \log c_t}_{\approx \Delta c_t / c_t} = \frac{1}{\sigma}\left[\log(1+r) + \log\beta\right] \approx \frac{r-\rho}{\sigma},$$

  using $\log(1+r) \approx r$ and $\log\beta = -\log(1+\rho) \approx -\rho$.

- **Continuous time**: the Euler equation is directly in growth rates,

$$\frac{\dot{c}_t}{c_t} = \frac{d}{dt}\log c_t = \frac{r_t - \rho}{\sigma}.$$

- For a general utility function, you can show the equivalence using a Taylor expansion (see Ch. 9.2 of the PhD Macrobook).

Frisch Elasticity: Derivation

← Back

- Taking logs of the labor supply condition (LS):

$$\log\phi - \log(1-\phi) + \log c_t - \log(1-\ell_t) = \log w_t$$

- Holding c_t constant (i.e., $\lambda_t = \bar{\lambda}$) and differentiating:

$$\frac{d\ell_t}{1-\ell_t} = d\log w_t \quad \Rightarrow \quad d\log(1-\ell_t) = -d\log w_t$$

- Using $d\log(1-\ell_t) = -\dfrac{\ell_t}{1-\ell_t}\, d\log \ell_t$:

$$-\frac{\ell_t}{1-\ell_t}\, d\log\ell_t = -d\log w_t \quad \Rightarrow \quad \left.\frac{d\log\ell_t}{d\log w_t}\right|_{\lambda_t=\bar{\lambda}} = \frac{1-\ell_t}{\ell_t} \equiv \varepsilon_F$$

- At the calibrated steady state $\ell^* = 1/3$: $\varepsilon_F = \dfrac{2/3}{1/3} = 2$.